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MILITARY SPECIFICATION

# BLOOD RECIPIENT SET, WITH METERING CHAMBER, WITHOUT NEEDLE, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a blood recipient set suitable for the indirect transfusion of whole blood and the administration of blood derivatives from bottles and standard blood collecting-dispensing bags in controlled volumes from 10 cubic centimeters (cc) to 100 cc.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| MMM-A-250 | Adhesive, Water-Resistant, (for Sealing Fiberboard Boxes). |
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |
| PPP-T-76 | Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing). |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |



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(Copies of specifications and standards required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. PHARMACOPEIAL CONVENTION, INCORPORATED

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

American National Standards Institute (A.N.S.I.)

ANSI Z70.1 Dimension of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

NATIONAL INSTITUTES OF HEALTH

Public Health Serivce Regulations - Part 73* Pertaining to Federal Control of the Sale, Barter, or Exchange of Biological Products.

*(Application for copies should be addressed to the Laboratory of Biologics Control, National Institutes of Health, Bethesda, Maryland 20014.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality, and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible and safe (nontoxic), when tested as specified in section 4. Those rubber and plastic components coming in contact with liquids being administered shall contain not more than 5 parts per million (p.p.m.) of lead when tested as specified in 4.4.2. The plastics comprising the fluid path shall meet the requirements of the Pharmacopeia of the United States (USP) for Class II plastics and for Transfusion and Infusion assemblies.



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ed necessary to assure supplies and services conform to inspections are deemed

3.1.1.1 Molded rubber part. The molded rubber part shall be fabricated from natural gum rubber latex.

3.1.1.2 Plastic tubing. Plastic tubing shall be clear, nontoxic, colorless, transparent, flexible polyvinyl, and shall meet the requirements for Class II plastics as stated in the USP. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be free of any particulate matter. The tubing shall be free from all liquid lubricating substances which oxidize, exude, or sweat-out to the surface of the tubing.

3.1.1.3 Piercing device. The piercing device shall be fabricated from a suitable rigid plastic.

3.1.1.4 Needle adapter. The needle adapter shall be fabricated from a suitable rigid plastic.

3.1.1.5 End protectors. End protectors shall be fabricated from plastic tubes or sheaths.

3.1.1.6 Medicinal Entry Assembly. The medicinal entry assembly shall be assembled from a part molded of a suitable plastic, a rubber diaphragm stopper, and a stopper securing band.

3.1.1.6.1 Rubber diaphragm stopper. The rubber diaphragm on the medicinal entry assembly leg shall be molded from natural, non-oxidizing, low-sulfur, wax-free rubber of 40-50° Shore A Durometer.

3.1.1.6.2 Diaphragm stopper securing band. The diaphragm securing band shall be fabricated of either latex rubber or heat-shrunk polyvinyl chloride.

3.1.1.7 Metering chamber. The metering chamber shall be fabricated of 10 gage polyvinyl chloride film. The film shall be free from all liquid lubricating substances which oxidize, exude or sweat-out to the surface of the film.

3.1.1.8 Filter chamber.

3.1.1.8.1 Input and output connectors. Input and output connectors shall be molded from suitable rigid plastic.

3.1.1.8.2 Barrel. The barrel shall be fabricated from a cut piece of clear, transparent, semi-rigid plastic tubing. The barrel shall be free from all liquid lubricating substances which oxidize, exude or sweat-out to the surface of the barrel.



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100 ml with purified water

3.1.1.8.3 Filter. The ring and framework of the filter shall be molded opaque white thermoplastic. The mesh shall be of nylon woven in a square lock crimped pattern.

3.1.1.9 Screw type control clamp. The screw type control clamp shall be fabricated from a suitable rigid plastic.

3.1.2 Metal slide clamp. The metal slide clamp shall be fabricated from aluminum. The clamp shall be compatible with the plastic tubing used to fabricate the upper solution input leg.

3.1.3 Other material. All other materials hereinafter specified shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Style and design. Style and design shall be in accordance with Figure 1 and as hereinafter specified.

3.2.1 Description. The blood recipient set shall be a disposable set consisting of a piercing device (coupler with piercing tip) a short length of transparent plastic tubing, a shut off clamp, a medicinal entry assembly, an in-line graduated metering chamber, a filter housing containing a nylon mesh filter, a long length of transparent plastic tubing, a molded rubber part, one flow control clamp, a needle adapter, end protectors and an instruction sheet. The set shall provide a sterile, pyrogen-free fluid path, and shall be suitable for the indirect transfusion of whole blood and the administration of blood derivatives from blood bottles and standard blood collecting-dispensing bags (NSN's 6515-00-926-4766 and 6515-00-926-4767). The set shall be adaptable, without modification, for connection to and operation with a blood warming coil (NSN 6515-00-926-2205).

3.2.1.1 Piercing device. A sharpened plastic piercing device (coupler with piercing tip) for the purpose of puncturing the blood container and providing a fluid path from the blood container to the recipient set, shall be firmly attached in a tight, leakproof manner to a 1-1/2 to 1-3/4 inch length of transparent plastic tubing. The piercing device shall be a truncated cylindrical tube, molded in one piece, 1-7/8 inches (plus or minus 1/8 inch) overall in length. The piercing device shall have a flat circular flange 3/4 inch (plus or minus 1/8 inch) in diameter, and a thickness of not less than 1/16 inch, permanently positioned on the long axis of the coupler. The flange shall be so constructed as to serve as the point to apply pressure to insert the piercing tip into the standard blood bag outlet ports and blood bottle stopper, and shall be of sufficient strength for this purpose. The penetrating end of the piercing device shall be 0.960 $\pm$ 0.010 inch in length and extend to the flange. The outside diameter of the penetrating end shall taper from 0.260 ($\pm$ .010 inch) at the distal end of the flange shoulder to 0.200 ($\pm$ .010 inch) at the heel of the aperture of the piercing tip. The piercing tip of the coupler shall be in the form of a sharp truncated cylinder. A single central orifice shall furnish the pathway for the fluid. The effective area of the orifice shall permit passage of a 0.125 inch (minimum) diameter stylet throughout the lumen of the piercing device. The piercing tip shall be sharpened to such an edge as to be suitable for the puncture of, and insertion into the port of a standard plastic blood bag, without damaging



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the bag, or into the blood bottle stopper shown in Figure 2, without exertion of undue pressure. The construction of the piercing device shall be such as to prevent the retention of clots or fibrin. The piercing device shall make a snug, leakproof fit with the blood bottle stopper and the outlet ports of the plastic blood bag for the duration of the use of the recipient set, and shall permit the recipient set to be removed from the stopper or ports of the bag without difficulty and without damage to the stopper or the bag (that is, except for the puncture). The piercing device shall be ruggedly designed and structurally sound. It shall not break, leak air or fluid during use. The piercing device shall not be brittle or soft, and shall not become brittle or soft under normal storage conditions.

3.2.1.1.1 Durability of piercing device. The durability of the piercing device shall be tested as specified in 4.4.9. The piercing device shall be so firmly attached to the transparent plastic tubing that the juncture of the device and the tubing shall not loosen or permit leaking nor shall the device or tubing become damaged. The piercing device shall be capable of being inserted into the blood containers (into the blood bottle rubber stoppers, Figure 2, and the outlet ports of the blood bag) as described in 4.4.8, without the device breaking or bending to the degree of causing difficulty or improper functioning when properly inserted into the blood container.

3.2.1.2 Plastic input tubing. The plastic tubing which connects the piercing device to the medicinal entry assembly shall be 1-1/2 to 1-3/4 inch length of transparent plastic tubing with a nominal inside diameter of 0.125. The plastic tubing shall be connected to the piercing device by insertion into the proximal (lower) end of the device and the plastic tubing shall be connected on the opposite end to the single leg of the medicinal entry device. The attachment shall be tight and leakproof when tested in accordance with 4.4.6. A captive metal slide clamp designed to close off the solution flow shall be positioned on this tubing. Tubing wall shall be of sufficient strength to withstand cutting when tested as specified in 4.4.9.

3.2.1.3 Metal slide clamp. The metal slide clamp shall be a rectangular piece of aluminum 1 inch ($\pm$ 1/8 inch) in length and 3/8 "($\pm$ 1/16 inch) in width, a nominal 0.025 inch thick with a nominal 13/16" longitudinal opening so tapered and slotted as to provide an open and closed position. The short ends of the clamp shall be bent over to provide for the application of finger pressure. The clamp shall be held captive on the plastic input tube between the piercing device and the medicinal entry assembly. The slot (opening) shall be of such dimensions that when the tubing is at the wider end of the taper (open position) the clamp is capable of being rotated 360° around the tube without binding or scoring the tube. When the tubing is forced into the narrow end of the slot (closed position), fluid flow shall be completely shut off. Exterior corners and edges of the clamp shall be slightly beveled to prevent the clamp from cutting the fingers during the application of pressure required to open or close the clamp. Edges and sides of the slot shall be slightly beveled and smooth enough to prevent the clamp from cutting the tube when in either the open or closed position. The action of opening and closing the clamp shall not cut the tubing when tested in accordance with 4.4.13.



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3.2.1.4 Medicinal entry assembly. The medicinal entry assembly shall be a molded Y shaped tubular device consisting of a fluid input leg opposed on the same plane by a fluid output leg and a medicinal entry leg. The fluid input leg shall be attached to the plastic input tube. The approximate overall length of the assembly shall be 1-1/2 inches. The medicinal entry leg shall have a female Luer taper on the proximal end and shall join with the fluid input leg in order to assure easy and rapid injection of medications. The appearance of the assembly shall be that of a Y with the medicinal entry leg in line with the output leg and the fluid input leg joining the two thus forming the characteristic Y shape. The minimum inside fluid passage diameter of the assembly shall be 0.093 inches.

3.2.1.4.1 Rubber diaphragm. The rubber diaphragm shall be sleeve style stopper with a tapered plug, diaphragm face and sleeve design which, when assembled to the medicinal entry leg of the medicinal entry assembly, allows both outside and inside circumference contact to assure secure retention of the diaphragm upon withdrawal of the medicinal needle and syringe. After puncture, the diaphragm face shall reseal without a falling drop within 10 second when tested as specified in 4.4.12.

3.2.1.4.2 Diaphragm securing band. The external sleeving of the diaphragm shall be further secured to the medicinal entry leg of the medicinal entry assembly by a diaphragm securing band forming an airtight, leakproof connection between the diaphragm sleeve and the medicinal entry leg which shall not separate or leak when tested as specified in 4.4.6.

3.2.1.5 Plastic output tubing. The plastic output tubing shall be 1-1/2 to 1-3/4 inches in length with a nominal inside diameter of 0.125 inches. The output tube shall securely connect the medicinal entry assembly to the bushing of the graduated metering chamber. The connection shall not leak or separate when tested as specified in 4.4.6. The tubing shall be free from liquid lubricating substances which oxidize, exude or sweat-out to the surface.

3.2.1.6 Metering chamber. The metering chamber shall be a bag fabricated by electronically sealing two transparent sheets of 10 gage polyvinyl chloride film with nominal inside dimensions of 2 x 5-11/64 inches and a volume capacity of 100 $\pm$ 5 cc. The chamber shall be connected to the plastic output tubing at the upper end and to the input connector of the filter chamber at the lower end. The chamber seals shall be a minimum of 1/16 inch wide and of sufficient strength to resist separation or leakage when tested as specified in 4.4.6. The end seals of the chamber shall be bonded to connector bushings at chamber input and output. The plastic film shall be free from liquid lubricating substances which oxidize, exude or sweat-out to the surface. The chamber shall be imprinted with a double volumetric scale graduated in 10 cc increments (0 through 100) indicating both the volume contained and the volume delivered. Volume contained shall be at the left hand side of the scale as one faces the chamber and the volume delivered at the right hand side. This scale shall be legibly imprinted with a non-toxic ink compounded to bond to the vinyl surface and assure against loss of legibility.



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3.2.1.7 Filter chamber. The filter chamber shall be constructed of an input connector, a barrel, a molded filter and an output connector. The chamber shall be flexible enough to permit squeezing by hand, yet rigid enough to withstand the collapse test as specified in 4.4.7. The chamber shall not leak or separate when tested as specified in 4.4.6. The filter chamber shall be so assembled that the portion within the barrel above the upper ring of the molded filter shall constitute a drip chamber. The drip device shall be so constructed that the fluid into the drip chamber shall be broken into drops, 12 such drops ± 3 drops, shall equal 1 ml. The filter chamber shall be constructed to prevent the passage of clots or fibrin. Chamber shall be constructed so that the fluid shall pass through the filter and through the output connector into the attached tubing.

3.2.1.7.1 Input connector. The input connector shall consist of a tapered portion approximately 1 inch long which shall be securely inserted into the fitting provided in the metering chamber, and flanged fitting securely connected to the barrel. The fluid passage which begins at the tapered portion above the flange shall extend below the flange for at least 3/8 inch and shall be visible through the barrel for approximately 1/16 inch below the collar of the flange to permit clear observation of the drop formation within the drip chamber. The fluid passage internal diameter shall be 0.091 inch (± 0.005 inch) at the tip of the tapered connector opening to 0.170 inch (± 0.005 inch) at the base of the port within the barrel.

3.2.1.7.2 Barrel. The barrel shall be 5 ± 1/8 inches long with a nominal diameter of 0.645 inch and a nominal wall thickness of 0.040 inch. The barrel shall be securely affixed to the input and output connectors and shall not leak when tested as specified in 4.4.6.

3.2.1.7.3 Filter. The molded filter shall consist of a nylon mesh insert molded into a ring and a frame to support the mesh. The ring shall fit tightly into the barrel and shall not permit fluid to pass between the exterior surface of the ring and the internal surface of the wall of the barrel. The overall filter length shall be 3-1/2 ± 1/8 inches. The nylon mesh shall be woven in a square lock crimped pattern of filaments 100 microns in diameter. The mesh count shall be 70 ± 5 per inch. Filter shall have a filtering area of not less than 3-1/2 square inches.

3.2.1.7.4 Output connector. The output connector shall consist of a tapered portion for insertion into the plastic tubing and a flange with collar for insertion into the barrel. It shall be so designed that it permits a secure fit into the barrel at one end and allows a proper tubing attachment at the other end. The connector will not leak or separate when tested as specified in 4.4.6. The minimum fluid passage shall be 0.075 inch diameter.



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3.2.1.8 Plastic tubing. Plastic tubing shall be 53 to 56-1/2 inches long with a nominal internal diameter of 0.133 inches and a nominal wall thickness of 0.022 inch. The plastic tubing shall extend from the output connector of the filter chamber to the molded rubber part. Connections shall not separate or leak when tested as specified in 4.4.6.

3.2.1.9 Molded rubber part. The molded latex rubber part shall be a minimum of 1-1/2 inches and a maximum of 2-1/4 inches in length, and shall be constructed with an enlarged center section approximately 5/8 inch in length, and a minimum wall thickness of 0.060 inch. The center section shall be shouldered on one end to provide for easy insertion of an 18 gage needle when adding supplemental medications. The molded rubber part shall be capable of resealing subsequent to puncture with a nonsiliconized 18 gage needle cannula. When tested for resealing as specified in 4.4.11, there shall be no falling drop within 10 seconds. The molded rubber part shall be connected to the plastic tubing and to the needle adapter.

3.2.1.10 Needle adapter. Needle adapter shall be fabricated of suitable rigid plastic and so constructed as to be attached at one end to the molded rubber part in a leakproof manner. The other end of the adapter shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI Standard Z70.1 in a tight leakproof manner (see 3.5 and 3.6). The adapter, or the assembly of the adapter to the molded part, shall be such as to prevent accidental reversal of the assembly of the adapter shall not separate from the molded rubber part when tested as specified in 4.4.6.

3.2.1.11 End protector. End protectors shall be readily removable plastic tubes or sheaths stuffed with non-absorbent, bleached cotton to permit passage of sterilizing gases. The end protectors shall maintain sterility, freedom from pyrogens and integrity of the entire fluid path of the infusion device until manually removed.

3.2.1.12 Control clamp. The control clamp shall be of the screw type, and shall be attached, in the open position, to the plastic tubing described in 3.2.1.8 and shall be movable on the tubing. The clamp shall provide a constant setting and quick shutoff from the operating position, and shall be capable of one-hand operation. To assure constant delivery of fluid, it shall be impossible for the initial clamp setting to be changed drastically by a sudden hard pull on the tubing. The control clamp shall not cut or damage the tubing in either the open or closed position. When tested as specified in 4.4.9, the clamp shall be capable of accomplishing complete quick shutoff (that is, no falling drop in drip chamber or from needle adapter), shall regulate flow of the fluid from complete shutoff to maximum flow, and with a constant head of water shall maintain a constant flow rate at any point between complete shutoff and maximum flow. In addition, when the clamp is operated from its fully open position to complete shutoff, and this operation is repeated ten times while the clamp is in the same position on the tubing, the tubing shall not be cut, pierced, or damaged so as to cause leaking or restriction of flow of fluid.



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3.2.1.13 Instruction sheet. Each set shall be provided with a detailed instruction sheet containing ample illustrations, and clear, concise instructions for proper use of the set with the standard plastic blood bags and a stoppered vacuum blood bottle. Instructions shall include a conversion table permitting an estimation of the number of milliliters infused per minute based on observed number of drops falling from the drip device in a 15-second interval. The table shall also include a method for computing the total quantity of blood infused in a given time (and vice versa).

3.2.2 Assembly. The needle adapter shall be connected to one end of the molded rubber part. The other end of the molded rubber part shall be connected to the plastic tubing (3.2.1.8). The filter chamber shall be connected at one end to the plastic tubing and at the other end to the metering chamber. The plastic output tubing (3.2.1.5) shall connect the metering chamber to the output leg of the medicinal entry assembly (3.2.1.4). The plastic input tubing (3.2.1.2) shall connect the input leg of the medicinal entry assembly to the piercing device and shall hold the metal slide clamp. The end protectors shall be fitted over the piercing device and the needle adapter.

3.3 Compatibility. The materials of which the sets are manufactured shall not react with, or be acted upon, by the liquids which they will be in contact with during normal use and operation, nor shall there be any materials incorporated in the set which will cause any harmful or deleterious effect when the set is used for the purpose intended.

3.4 Sterilization, toxicity, and freedom from pyrogen. All equipment shall be nontoxic (safe). Fluid paths, end connectors and airway cannulae of sets shall be sterile and pyrogen-free. All of the above shall be in accordance with the additional standards: Whole Blood (Human) of the Public Health Service Regulations - Part 73. The fluid path and end connectors shall be pyrogen-free when tested as specified in 4.4.5, and sterile when treated as specified in 4.4.3. The sets shall be non-toxic (safe) when tested for safety as specified in 4.4.4.

3.4.1 Certificates. For destination inspection only, certificates of sterility, freedom from pyrogens, and safety, as specified below, shall be submitted by the contractor directly to Technical Operations Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101, Attention: Quality Assurance Branch, and one copy of each shall be forwarded to the consignee.

3.4.1.1 Certificate of sterility. A certificate of sterility, listing all sterility lot or control number, shall state that samples from each sterility lot or control have been tested for sterility, as required in 3.4, and were found to be sterile. The certificate shall reference the method used in determining sterility.



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3.4.1.2 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety, as specified in 3.4, and were found to comply with such requirements. The certificate shall reference the methods used in determining safety.

3.4.1.3 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens, as specified in 3.4, and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.5 Leakage of set. Sets shall be tested for leakage as specified in 4.4.6. Sets which show leakage, loosening, or separation of any part of the set shall be rejected.

3.6 Separation of set. The piercing device or the recipient tube shall not separate, or loosen, from the filter-drip housing when tested as specified in 4.4.6.

3.7 Filter and drip chamber collapse. Sets shall be tested for filter and drip chamber collapse as specified in 4.4.7. Sets in which filter or drip chamber collapses sufficiently to empty fluid from drip chamber and allows air to enter recipient tube shall be rejected.

3.8 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made on nontoxic, sterile, pyrogen-free sets. These records shall be made available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, or any official of the Federal Food and Drug Administration, or any duly authorized Government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.9 Government furnished material. The Government shall furnish the contractor the plastic blood bags (NSN 6515-00-926-4767) needed to perform the piercing device durability test specified in 4.4.8 and the clamping device specified in 4.4.9.

3.10 Workmanship. Workmanship shall be first class throughout. All materials, treatments, and procedures used in the fabrication and assembly of the recipient sets shall be such that the finished sets shall not show any evidence of decomposition, deterioration, or other conditions which may adversely affect the appearance or serviceability of the sets at time of delivery or during storage.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the rubber or plastic, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination purposes shall be one blood recipient set.

Table I. Sampling for Examination

| Type Defect | Inspection Level | AQL (percent defective) |
|---|---|---|
| Major | II | 1.5 |
| Minor | II | 2.5 |



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4.2.2 For tests.

4.2.2.1 For sterility test. Sampling for the sterility test shall be in accordance with USP for parenteral devices.

4.2.2.2 For pyrogen, and safety tests. Sampling procedures for pyrogen, and safety tests shall be conducted as indicated in the USP, for transfusion and infusion assemblies.

4.2.2.3 For lead test. Samples shall be taken from each lot of rubber or plastic as indicated in paragraph 4.4.2.

4.2.2.4 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

Table II. Sampling for other Tests

| Component (unit of product) | Characteristic | Req't | Test Procedure | Insp. level | AQL (percent defective) |
|---|---|---|---|---|---|
| 1-Blood Recipient Set | Needle adapter | 3.2.1.10 | 4.4.1 and 4.4.6 | S-2 | * |
| 1-Blood Recipient Set | Leaking of set | 3.2 | 4.4.6 | S-2 | * |
| 1-Blood Recipient Set | Filter & drip Chamber collapse | 3.2.1.7 | 4.4.7 | S-2 | * |
| 1-Blood Recipient Set | Piercing device durability | 3.2.1.1.1 | 4.4.8 | S-2 | * |
| 1-Blood Recipient Set | Clamping device | 3.2.1.3 3.2.1.12 | 4.4.9 | S-2 | * |
| 1-Blood Recipient Set | Reseal test (molded rubber part) | 3.2.1.9 | 4.4.10 | S-2 | * |
| Plastic components | Material-Class II | 3.1.1 | 4.4.11 | ** | * |
| 1-Blood Recipient Set | Reseal test (diaphragm) | 3.2.1.4.1 | 4.4.12 | S-2 | * |
| 1-Blood Recipient | Metal slide clamp | 3.2.1.3 | 4.4.13 | S-2 | * |



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* Acceptance number shall be zero for all sample sizes. Minimum of two samples shall be taken regardless of lot size.

** Sufficient sample units to perform the biological tests shall be selected from each lot.

4.2.2.5 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The blood recipient sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any sets in the sample containing one or more defects shall be rejected and if the number of defective sets in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects in Accordance with MIL-STD-105

| Categories | Defects |
|---|---|
| Critical | None defined: |
| Major | |
| 101 | End protectors missing or dislodged. |
| 102 | Closed type protector not free of holes, or cotton missing (or inadequate amount) from open type protector. |
| 103 | Tubing and filter housing not free of bloom or tackiness. |
| 104 | Set incomplete. |
| 105 | Style and design not as specified. |
| Minor | |
| 201 | Dimensions not as specified. |
| 202 | Contractor's name or registered trademark, missing, incomplete, or illegible. |
| 203 | Instructions missing, incomplete, or illegible. |

4.4 Tests.

4.4.1 Needle adapter. The tapered end of the needle adapter shall be tested for proper dimensions and taper by means of the ring gage shown in Figure 7 of A.N.S.I. Standard Z70.1. This gage shall be slipped onto the tip with moderate pressure.



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4.4.2 Test for lead. Ten (10) gms. of rubber or plastic shall be taken, ignited to char, and then further ignited in a muffler at 700° C. to a white ash. The ash shall be boiled in 15 ml of 10 percent nitric acid filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the USP, shall be applied using a control of 5 p.p.m. or 12 p.p.m. of lead solution as given in the USP test. If the sample solution is darker than the limit, smaller aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in p.p.m. is determined.

4.4.3 Sterility. Samples of the finished recipient sets shall be tested for sterility, in accordance with the procedures for parenteral devices as described in the USP.

4.4.4 Safety (toxicity) test. Samples of the finished sets shall be tested for safety, in accordance with the USP for transfusion and infusion assemblies.

4.4.5 Freedom from pyrogens. Samples of the finished sets shall be tested for freedom from pyrogens, in accordance with the USP for transfusion and infusion assemblies.

4.4.6 Leakage of set test. The set shall be subjected to an internal air pressure of 4 pounds per square inch gage (psig) while immersed in a water bath. No rising bubbles shall be noted in the 10 seconds following application of the pressure.

4.4.7 Filter and drip chamber collapse test. The filter chamber shall be filled to one-quarter its capacity, the drip chamber to one-third its capacity, and the recipient tube shall be completely filled with water at 100° F. Close off the entrance to the piercing connector lumen. Hold set in a vertical position, with needle adapter three feet below the bottom of the drip chamber. Remove end protector from needle adapter and allow fluid to flow from open end of set. Subject to this condition for one minute.

4.4.8 Piercing device durability test. The piercing device (on assembled set) shall be inserted into, and removed from, a blood bottle stopper as shown on Figure 2 and inserted and twisted into, and removed from, a plastic blood bag (NSN 6515-00-926-4767, ), in the normal manner. This operation shall be repeated twice using a total of three bottles and three bags. The piercing device shall then be immersed in Anticoagulant Solution "A" in accordance with the USP and the Additional Standards: Whole Blood (Human) of the Public Health Service Regulations - Part 73, for 4 hours. The piercing device shall then be removed from the solution, and immediately inserted in a plastic blood bag as specified above.



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4.4.9 Clamping device test. The assembled set shall be connected, in the normal manner, to a plastic blood bag (see 3.10) which has been filled with water and sealed against leaking. The blood bag, with set connected thereto, shall be suspended in the normal manner, at a sufficient height to permit the entire length of the tubing of the set to hang vertically. The ability of the clamp to accomplish complete shut off, to regulate flow, and, with a constant head of water, to maintain a constant rate to flow, shall be tested with the clamp at a point on the plastic tubing within two inches above the molded rubber part. The tests shall be performed with the tubing hanging vertically at all times. The rate of flow shall be determined either by counting the drops in the drip chamber during a measured time period, or by recording the time required to collect a measured volume of fluid from the needle adapter.

4.4.10 Reseal test (molded rubber part). This test shall be applied to those samples previously subjected to the leakage test (4.4.6). The assembled set shall be connected to a full, suitably vented, one liter bottle of water, saline solution of dextrose solution. The bottle shall be hung in a normal manner so that the liquid level in the bottle shall be three feet above the injection site, and the set shall be adjusted to a drip rate of 25 drops (plus or minus 2 drops) in 15 seconds. The test shall be completed before the bottle is one half empty. While the solution is flowing, the molded rubber part shall be placed in a horizontal position and shall be punctured from the underside in the shouldered section with an 18 gage non-siliconized needle and the needle immediately withdrawn. After puncture the molded rubber part shall meet the requirements specified (see 3.2.1.9).

4.4.11 Biological tests. The plastics comprising the fluid path shall be tested in accordance with the USP for Class II plastics.

4.4.12 Reseal test (rubber diaphragm). The rubber diaphragm shall reseal without a falling drop within 10 seconds after puncture with a nonlubricated 18 gage needle when subjected to an internal pressure (hydrostatic pressure) of 15 p.s.i.g. for 30 seconds.

4.4.13 Metal clamp test. The metal slide clamp shall be operated for at least 10 times to stop the flow and an equal number of times to allow full flow when subjected to an internal pressure (hydrostatic pressure) of 4 p.s.i.g. At the completion of the test the tubing shall be examined for cuts.



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## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One blood recipient set, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 12 units | 96 units |

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each set shall be packaged in a box of appropriate size and design in such a manner as to prevent kinking of the tubing. Commercial colors are acceptable on the box and closure shall be adequate to prevent spilling of contents under normal handling. Each box shall be further enclosed in a transparent watervaporproof barrier and heat sealed.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.



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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129 and the requirements of the Federal, Food, Drug, and Cosmetic Act. Date of manufacture and lot (control) number shall be shown. In addition, the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-9/69-G". Final sterilization shall be on or after indicated packaging date. Marking shall include the legend: "FLUID PATH AND END CONNECTORS ARE STERILE AND PYROGEN FREE".

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name of contractor shall be shown. The method of sterilization as specified in 5.3.1 and the date of manufacture shall also be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number, date of manufacture and the method of sterilization as specified in 5.3.1 shall also be shown.

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6. NOTES

6.1 Intended use. The blood recipient set is intended for accurate administration of blood or blood derivatives from blood bottles and standard blood bags in controlled volumes from 10cc to 100cc.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog: NSN 6515-00-958-7795, BLOOD RECIPIENT SET, INDIRECT TRANSFUSION, with Graduated Metering Chamber, Disposable. It does not include all types, sizes, or grades as indicated by the title or all types which are commercially available.

Custodians:
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Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

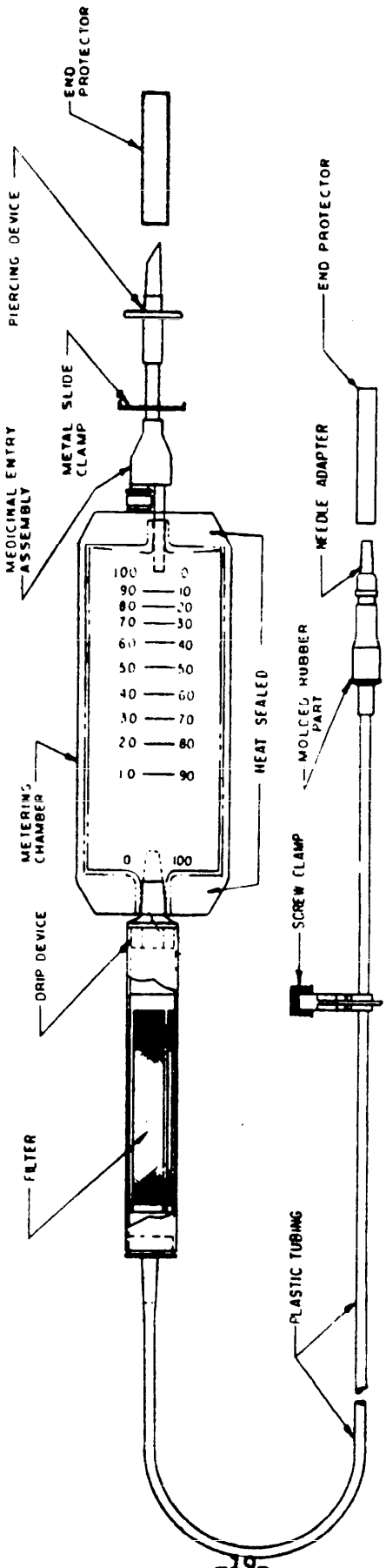
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STYLE DESIGN AND ASSEMBLY OF SET PROPER

FIGURE 1



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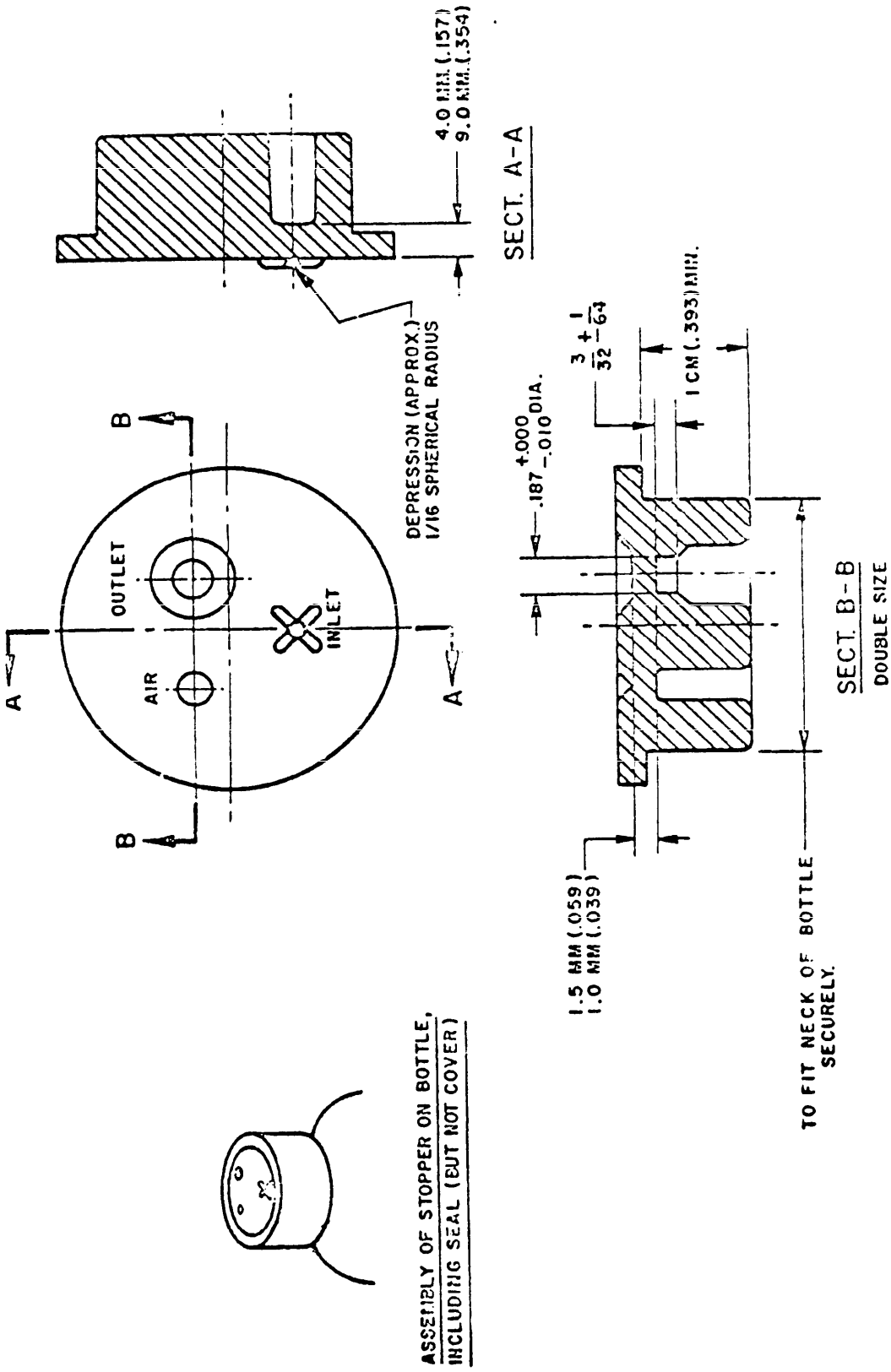



FIG. 2



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-B-36646A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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